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Longitudinal Comparison of Geographic Atrophy Enlargement Using Manual, Semiautomated, and Deep Learning Approaches

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Objective: To compare a fully automated artificial intelligence (AI) model, a semiautomated method, and manual planimetry in the longitudinal assessment of geographic atrophy (GA) using fundus autofluorescence images.

Design: A retrospective analysis of 3 GA assessment methods: AI, Heidelberg Eye Explorer semiautomated software (RegionFinder), and manual planimetry.

Subjects and Controls: One hundred eight patients (185 eyes) with GA from a phase IIb clinical trial by GlaxoSmithKline, which evaluated an experimental drug that did not reduce GA enlargement compared with the placebo.

Methods: Fundus autofluorescence images of 185 eyes were annotated using manual planimetry, semiautomated RegionFinder, and a fully automated AI model trained and validated on manual planimetry annotations at screening, year 1, and year 2. Artificial intelligence masks were compared with human-guided methods, and regression errors were assessed by stacking masks from consecutive visits. Agreement between methods was assessed using Bland–Altman plots, Dice similarity coefficient (DSC), and comparisons of GA growth rates. Artificial intelligence performance was evaluated based on its need for human edits and frequency of regression errors.

Main Outcome Measures: Agreement between methods was evaluated using Bland–Altman plots, DSC, and intraclass correlation coefficients (ICCs). The mean GA growth rate (mm^2/year) and square root transformation of GA size were compared across methods. Artificial intelligence performance was assessed by the percentage of acceptable masks and the frequency of longitudinal regression errors.

Results: At screening, the mean GA area was 7.22 mm^2 with RegionFinder, 8.37 mm^2 with AI, and 8.66 mm^2 with manual planimetry. RegionFinder measured smaller GA areas than both AI and manual, with a mean difference of −1.45 mm^2 (95% confidence interval [CI]: −1.56, −1.35) versus AI (ICC = 0.945) and −1.87 mm^2 (95% CI: −1.99, −1.75) versus manual (ICC = 0.920). Growth rates were comparable between RegionFinder (1.54 mm^2/year), AI (1.68 mm^2/year), and manual (1.80 mm^2/year) ($P = 0.25$). Artificial intelligence masks were deemed acceptable in 84.8% of visits, and 81.4% of cases showed no regression over time.

Conclusions: Artificial intelligence accurately measures GA in approximately 85% of cases, requiring human intervention in only 15%, indicating potential to streamline GA measurement in clinical trials while maintaining human oversight.

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Age-related macular degeneration is a leading cause of severe vision impairment in older adults.[1,2] Geographic atrophy (GA) is an advanced stage of age-related macular degeneration characterized by loss of photoreceptors, retinal pigment epithelium, and choriocapillaris.[2] Areas of GA produce dense scotomas, which can progress to involve the fovea and cause central vision loss.[1,2] Fundus autofluorescence (FAF) imaging is a vital component utilized in the multimodal assessment of GA.[2,3] The loss of lipofuscin within the retinal pigment epithelium of GA results in a reduction in autofluorescent signal, or hypoautofluorescence (hypoAF), which can be delineated and quantified to determine the extent of disease.[4] Consequently, the FAF-derived change in anatomical area over time (enlargement rate) is recognized by the US Food and Drug Administration as a primary end point in clinical trials.[5,6]

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Measurement of GA area for clinical trial end points is frequently performed by reading center graders using a semiautomated software called RegionFinder (Heidelberg Engineering). This software requires FAF images to be acquired using the Heidelberg Spectralis system. Measurement with RegionFinder is preferred due to having an automatic area output, precise longitudinal tracking, and high intraobserver and interobserver reliability.[7–11] Despite its advantages, timely assessment is challenging because the algorithm uses grayscale intensity to highlight each lesion, requiring graders to carefully distinguish GA from intersecting areas of low contrast and physiologic areas of hypoAF, such as the fovea and vessels.[8–10,12] An alternate method is manual planimetry, in which the grader traces the borders of each lesion freehand. This method is not restricted to Spectralis FAF imaging systems and can be expanded to other FAF cameras. Various software programs and image file types can be used, making it a widely available and cost-effective option; however, the process is time-consuming and lacks longitudinal tracking, making it less reproducible than RegionFinder.[9] Ensuring accurate and reproducible quantitative assessment of FAF is critical to the success of clinical trials, yet the use of different image storage formats, segments, and measurement conversion processes results in a lack of uniformity among methods.[8,13]

The limitations of both manual and semiautomated techniques led to the development of various artificial intelligence (AI) algorithms, which can segment GA in a matter of seconds with good accuracy.[13,14] A key difference between AI and human-graded methods is the ability to access previous visits to ensure precise longitudinal tracking of GA. Despite the impressive performance reported by the various AI models in cross-sectional data, studies evaluating longitudinal performance of AI models are limited.[14–16] Considering its role as an important end point in clinical trials, establishing an AI tool that can longitudinally track GA at the level of human graders represents a critical step in its acceptance in clinical trials and in practice.

The goal of this study was to evaluate the methodologic agreement of AI, manual planimetry, and the RegionFinder software in the quantification of GA area. Specifically, the performance of these 3 methods in assessing GA enlargement over time was evaluated and compared.

Methods

Dataset

Fundus autofluorescence images with GA were obtained from a phase IIa clinical trial conducted by GlaxoSmithKline between 2011 and 2016 (NCT01342926). This was a multicenter study conducted across 40 centers in the United States and Canada in which the experimental drug did not slow the enlargement rate of GA compared with the placebo.[17] Fundus autofluorescence images were captured on Heidelberg Retinal Angiograph by certified photographers. A single image was acquired at 30° centered on the macula, captured in high-speed mode (768 × 768 pixels) using the automated real-time mean function set at 14, and exported to the reading center in the Heidelberg proprietary e2e format. Inclusion criteria required well-demarcated GA with an area of 1.9 to 17 mm^2 measured on color fundus photographs of the study eye. For multifocal GA, ≥1 of the foci had to be >1.9 mm^2 and the total area of GA had to measure <17 mm^2. The study was conducted under institutional review board approval at each site, and written informed consent was obtained from all study participants. The research followed the tenets of the Declaration of Helsinki and complied with the Health Insurance Portability and Accountability Act.

Annotation Methods

Fundus autofluorescence images from the GlaxoSmithKline clinical trial were segmented manually, semiautomatically, and with an AI model at screening (185 eyes and 108 subjects), year 1 (185 eyes and 108 subjects), and year 2 (84 eyes and 50 subjects). A total of 2 images were deemed ungradable and excluded from this study due to GA being indistinguishable from peripapillary atrophy. Certified graders employed standardized criteria to identify and measure GA. Hypoautofluorescence or GA was classified as well-defined, homogeneously black areas with a minimum size of 250 microns in its widest diameter or an area >0.05 mm^2. Images were split between 3 independent graders (C.P. and J.B.) with a random subset undergoing repeat grades for intergrader agreement. All images were annotated independently by graders using RegionFinder and manual planimetry. In addition, an AI model was also used to generate masks as described below. Figure 1 shows an example of GA segmentation using RegionFinder, AI, and manual planimetry.

RegionFinder Semiautomated Method. RegionFinder (version 2.6.1; Heidelberg Engineering) was used to detect and quantify GA semiautomatically in the same images coinciding with previously described methods.[8] Graders began by selecting a seed point corresponding to the densest region of hypoAF of each lesion, which was expanded based on manual adjustment of grayscale threshold until the GA borders were precisely demarcated. As areas of physiologic hypoAF often intersect with GA (e.g., macula and vessels), graders utilized a variety of constraint tools to ensure true borders of GA were demarcated. For each eye, the screening image was annotated and copied onto the year 1 image, then adjusted accordingly. This process was repeated for year 2 using the year 1 annotation to ensure precise longitudinal tracking.

OCT was not available for confirming foveal involvement. As an adjunct to standard reading center practices, graders used near-infrared images to aid in delineating GA from the fovea. Total area was automatically calculated based on the software's proprietary algorithm.

AI Model. Details of the deep learning model were previously published.[18] In brief, the U-Net model was trained on 601 FAF images from the Age-Related Eye Disease Study 2. Model performance on 601 images from the validation data showed a mean (standard deviation) difference of GA in mm^2 of −0.07 (−1.61 to 1.47) and a Dice similarity coefficient (DSC) of 0.885. In addition, the model was tested on an independent clinical dataset of 156 FAF images and had a mean difference of −0.24 (−4.98 to 4.49) with a DSC of 0.918. Each AI mask area was calculated using the scale from the original image stored in Heidelberg Eye Explorer.

All AI-generated masks were randomly assigned to 2 graders (T.F.S. and J.B.) for accuracy review and classified as acceptable or unacceptable.[19] To minimize bias, the assessment was conducted several months after the original annotations. Additionally, half of the masks were reviewed by an independent grader (T.F.S.).

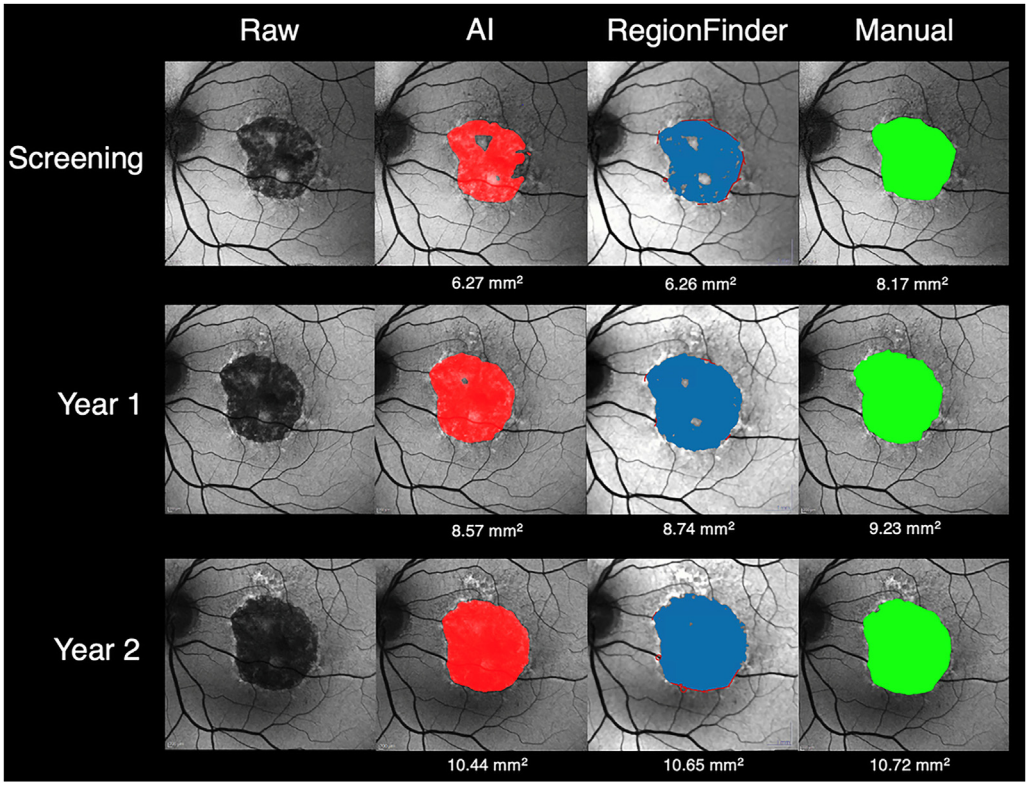


Figure 1. Measurement of GA area over 2 years with 3 methods: RegionFinder, AI, and manual perimetry. The GA is heterogeneous, as seen by lighter areas within the hypo-autofluorescent lesion. Both RegionFinder and AI methods exclude these lighter areas, and the manual method includes them resulting in a larger area and slower enlargement. AI = artificial intelligence; GA = geographic atrophy.

Acceptable masks included excellent segmentation with no edits or minor edits to the masks needed. Unacceptable masks were marked when critical errors were identified, including incorrect segmentation and errors in foveal involvement or peripapillary involvement. No edits were made to the AI-generated masks. Figure S1 (available at www.ophthalmologyscience.org) shows examples of acceptable and unacceptable segmentations.

Another area where AI-generated masks require supervision is longitudinal regression. While RegionFinder and manual annotation enable graders to compare previous visits for longitudinal context, recognizing that GA lesions can remain stable or enlarge over time, deep learning algorithms lack this capability unless additional postprocessing tools are added. As a result, the standalone AI model evaluates each visit independently, without reference to prior segmentations. This can result in observed variations where lesions may both increase and decrease over time with AI-generated masks. To assess for the presence of possible lesion regression, AI-generated masks of the screen and year 1 of each eye were stacked and color-coded; yellow indicated a consistent area at both visits, red indicated regression, and green indicated growth (Fig 2). Graders reviewed the longitudinal comparison and scored it as no regression, minor regression, and major regression. Minor regression was documented if <10% of a lesion appeared red or regression was confined to a single lesion. Major regression was documented if >10% of a lesion exhibited regression across 1 or multiple foci. Major regression was also considered in multifocal GA, where a lesion was annotated at 1 visit and not marked at all at the next visit, even if the lesion was small and accounted for <10% of area. Figure 2 shows examples of no regression, minor regression, and major regression.

Manual Planimetry. For manual planimetry, FAF images were annotated using the region overlay tool built into the Heidelberg Eye Explorer software. Graders traced each region of atrophy freehand, and then selected the fill area option to produce a solid overlay. In multifocal patterns of atrophy, each lesion area in mm^2 was manually summated using the software-generated output to ascertain the total area. Graders had access to previous visit masks while performing follow-up planimetry. All follow-up visits were graded by a single grader at the same sitting.

Statistical Analysis

The agreement between all 3 methods was assessed with the difference in mean GA area and DSC. Using RegionFinder as the gold standard comparison, the agreement between GA areas across all visits was assessed using Bland–Altman and scatter plots with corresponding intraclass correlation coefficients (ICCs). Dice similarity coefficient, a spatial overlap index, and a reproducibility validation metric were generated by comparing the 3 segmentation methods from the same visit and averaged at each time point. Comparison of longitudinal area measurements was performed using GA growth rate and change in square root GA lesion size from baseline to year 2. Square root transformation was calculated using previously described methods to adjust for baseline lesion size.[20] Comparison between annual growth rate and square root transformation of all 3 methods was performed using 1-way analysis of variance. Intergrader agreement for both human-guided methods was assessed with ICC and mean difference with 95% confidence interval (CI) graphed on Bland–Altman plots.

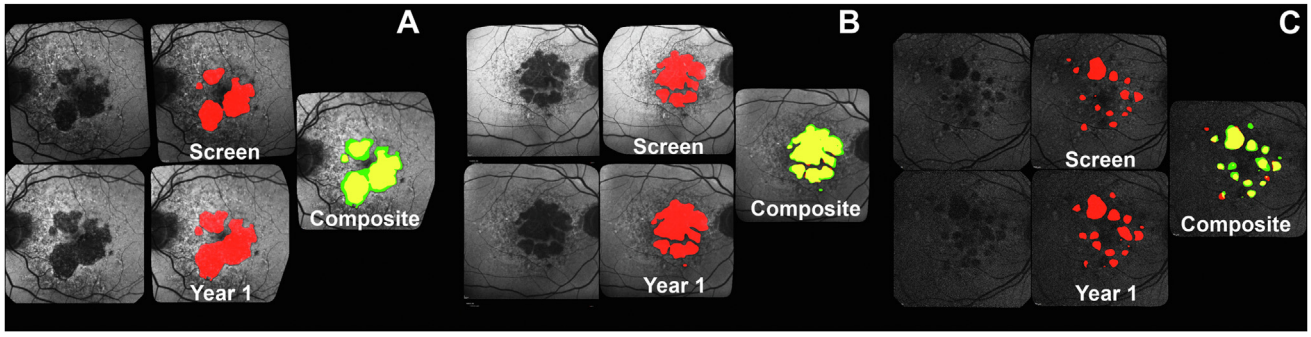


Figure 2. Analysis of longitudinal regression. The top row shows the raw image and corresponding AI mask at screening, while the bottom row displays the raw image and AI mask at year 1. The rightmost composite image represents the superimposed masks from both visits, where yellow indicates consistent GA area, red signifies regression, and green represents growth. Examples are provided for each grading category: no regression **(A)**, minor regression **(B)**, and major regression **(C)**. In panel C, the follow-up image misses GA foci due to suboptimal image quality. Instances where GA foci were missed at year 1 were classified as major regressions. AI = artificial intelligence; GA = geographic atrophy.

Results

Geographic atrophy area was measured in 185 eyes (108 participants) at screening and year 1 and 84 eyes (50 participants) at year 2. At the screening visit, GA was multifocal in 21% and subfoveal in 63% with a mean area of 7.22 (4.58) mm^2 with RegionFinder, 8.37 (4.98) mm^2 with AI, and 8.66 (5.21) mm^2 with manual planimetry.

Comparison of Area of GA

The mean area measurement at screening, year 1, and year 2 are shown in Table 1. Bland–Altman plots in Figure 3 compare area between RegionFinder and AI and between RegionFinder and manual planimetry. The mean GA area and difference between methods were calculated using measurements from all 454 images at all visits. Overall, RegionFinder was found to measure GA smaller than both AI and manual, with a mean difference between RegionFinder and AI of −1.45 mm^2 (95% CI, −1.56, −1.35); ICC = 0.945 and between RegionFinder and manual was −1.87 mm^2 (95% CI, −1.99, −1.75); ICC = 0.920. The DSC for comparison of masks between RegionFinder and AI was 0.923 and between RegionFinder and manual planimetry was 0.938. Artificial intelligence was similar to manual, with a mean difference of −0.42 mm^2 (95% CI, −1.98, 1.14); ICC = 0.989.

Intergrader agreement was assessed in 47 and 30 eyes by repeat segmentation using RegionFinder and manual planimetry, respectively, as shown in Figure S2 (available at www.ophthalmologyscience.org). The mean difference between graders using RegionFinder was 0.34 mm^2 (95% CI, −1.55 to 2.23) and manual planimetry was 0.36 mm^2 (95% CI, −1.03 to 1.75).

Progression Comparison

The mean annual growth rate and square root transformed growth rates are shown in Table 1. The mean GA growth over 2 years using the 3 methods is plotted in Figure 4. There was no statistically significant difference between growth rate (P = 0.25) measured using RegionFinder (1.54 mm^2/year), AI (1.68 mm^2/year), and manual (1.80 mm^2/year). Difference in mean square root transformed growth rate was also statistically insignificant (P = 0.38) between RegionFinder (0.84 mm/year), AI (0.87 mm/year), and manual (0.90 mm/year). The mean difference in growth rates between AI and RegionFinder were 0.26 mm^2/year (95% CI, −1.14 to 1.66) and 0.42 mm^2/year (95% CI, −1.17 to 2.01) between manual and RegionFinder. Individual eye-level variability is evident in the Bland–Altman plots (Figure S3, available at www.ophthalmologyscience.org). The ICC values indicate good agreement (0.807 for AI vs. RegionFinder, 0.746 for manual vs. RegionFinder), but method-dependent differences emerge at different growth rates. The differences tend to be larger at higher GA growth rates, suggesting that the segmentation methods diverge more in rapidly progressing eyes.

AI-Generated Mask Review

Grader assessment of AI-generated masks showed acceptable masks in 84.8%. Acceptability was done independent of follow-up visits and indicated that AI-generated masks did not need further edits for that visit. Assessment of color-coded masks for longitudinal regression showed acceptable longitudinal annotations with no regression in 81.4%. Of those that demonstrated regression, 11.3% had minor regression and 7.2% had major regression.

Discussion

This study compared GA measurements from RegionFinder, manual planimetry, and AI-based methods, with a focus on longitudinal data and growth performance. Our results showed that RegionFinder consistently measured smaller areas compared with manual measurements, which tended to be the largest, while AI-based measurements fell

Table 1. Mean GA Area, DSC at Each Time Interval, and Growth Rate for All 3 Methods

Cross-sectional AI Performance	Screen n = 185	Year 1 n = 185	Year 2 n = 84	*P* Value* among Slopes for 3 Methods
RegionFinder mean (SD) area (mm^2)	8.01 (4.78)	9.56 (5.19)	10.3 (5.38)	-
Manual planimetry mean (SD) area (mm^2)	9.64 (5.63)	11.6 (6.36)	12.3 (6.20)	-
AI mean (SD) area (mm^2)	9.32 (5.36)	11.2 (6.05)	11.7 (5.94)	-
DSC-manual planimetry to Region Finder	0.936	0.940	0.950	-
DSC-AI to RegionFinder	0.923	0.928	0.939	-

Longitudinal AI Performance	RegionFinder	Manual planimetry	AI	
GA growth rate (mm^2/yr)	1.54	1.68	1.80	0.25
Sqrt GA growth rate (mm)	0.84	0.87	0.90	0.38

AI = artificial intelligence; DSC = Dice similarity coefficient; GA = geographic atrophy; SD = standard deviation; Sqrt = square root transformation.
*One-way analysis of variance test.

between the 2. Despite these differences, no significant variation in growth rates was observed across the 3 methods.

Consistent with the previous work, RegionFinder was shown to regularly measure GA smaller than other methods.[8,9,11,21] As suggested by Mahmoudzadeh et al,[9] this is primarily thought to be related to its reliance on grayscale intensity to classify GA; heterogeneous lesions (Fig 1) or larger lesion areas, as shown in Figure S4 (available at www.ophthalmologyscience.org), were consistently measured larger using manual planimetry and the AI, likely because these methods tend to include a wider range of hypoAF intensities within each lesion. This is evident in our Bland–Altman plots, where small lesions show minimal difference between the methods, but as lesion size increases, the difference becomes more pronounced, with RegionFinder measuring significantly smaller areas compared with manual and AI methods. Interestingly, the AI model in this study was trained on manual planimetry annotations, yet it will often exclude small areas of isoautofluorescence within heterogeneous lesions, similar to RegionFinder (Fig 1).

Precise longitudinal GA growth tracking is critical in clinical trials and practice.

This study used both a quantitative and qualitative assessment to evaluate the performance of AI and the most commonly used techniques. Although the AI model used in our study lacks the longitudinal context that human graders rely on to ensure consistent tracking of GA growth, we show that the GA growth rate is comparable between each method. The decision to provide the AI model with no longitudinal context was intentional so we could assess whether the model was robust enough to accurately segment without the need for postprocessing or a reliance on past annotations. We specifically examined this lack of temporal framework in AI by examining baseline and follow-up masks overlayed and color-coded for enlargement and regression. We found regression in about 18% of eyes, with significant regression in 7%. These are the types of errors that a human grader with temporal context would likely avoid, such as missing a lesion previously identified or inconsistently including the fovea at baseline but not during follow-up. Some studies have used fully automated AI measurements for GA growth monitoring and could miss these errors.[22,23] Exploring time-series analysis capabilities with model development can improve the accuracy of longitudinal tracking.

While AI could be used in a semiautomated capacity, similar to RegionFinder, allowing for human edits to refine segmentations, our focus was on evaluating its fully automated capabilities and standalone potential, particularly in clinical workflows where human intervention may be minimal or impractical. Fully autonomous AI could

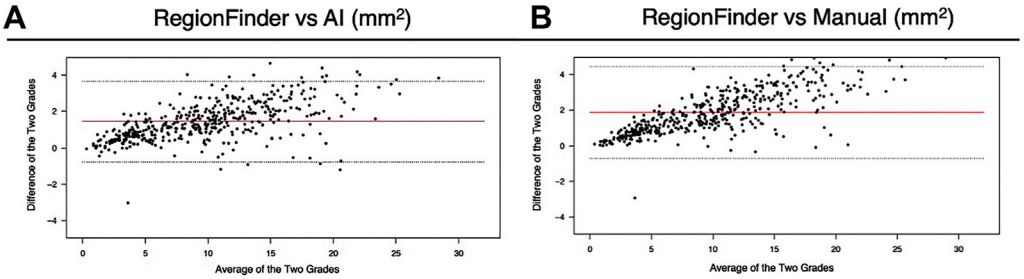


Figure 3. Bland–Altman plots for comparison of the GA area between RegionFinder and AI **(A)** and RegionFinder and manual planimetry **(B)**. AI = artificial intelligence; GA = geographic atrophy.

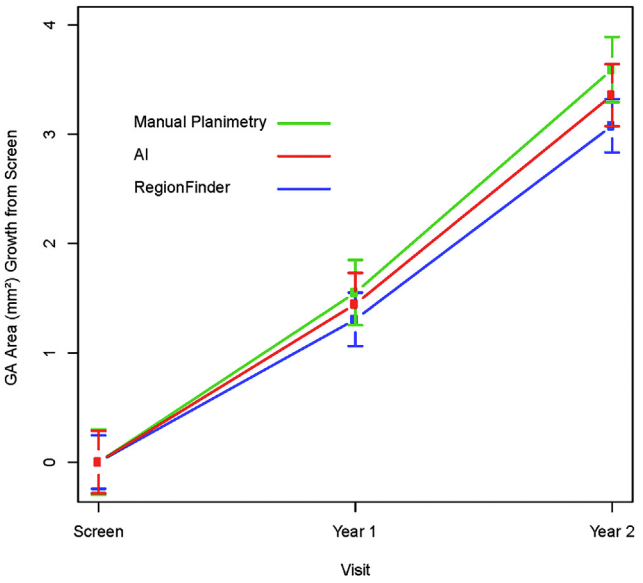


Figure 4. Comparison of change in area of GA over 2 years as measured by RegionFinder, AI, and manual planimetry. Mean growth rate was 1.54 mm^2/year, 1.68 mm^2/year, and 1.80 mm^2/year, respectively. AI = artificial intelligence; GA = geographic atrophy.

significantly reduce clinician workload, allowing for quicker, more standardized assessments. While area of GA comparison and DSC are excellent statistical measures for validating AI, another practical method is acceptance by a human expert. Our grader assessment showed that the AI mask would be acceptable as is 85% of the time. This implies that to meet the quality standards expected in clinical trials, it will be essential to pair AI segmentation models with tools that allow edits to the masks by human experts. One key advantage of AI is that only a small percentage of cases (15%) required edits, compared with RegionFinder, which typically requires human edits for every case. Future studies could explore the utility of AI as a semiautomated tool, where human operators review and edit AI-generated masks, similar to how RegionFinder is used.

In conclusion, while RegionFinder consistently measured smaller areas and manual methods measured larger, all methods provided reliable longitudinal growth tracking. Although AI performed well in its fully automated capacity, future work should explore AI as a semiautomated tool, allowing for human intervention to improve segmentation accuracy in clinical practice.

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Analysis and interpretation: Bogost, Slater, Voland, Domalpally

Obtained funding: N/A

Overall responsibility: Bogost, Domalpally

Abbreviations and Acronyms:
AI = artificial intelligence; **CI** = confidence interval; **DSC** = Dice similarity coefficient; **FAF** = fundus autofluorescence; **GA** = geographic atrophy; **HypoAF** = hypoautofluorescence; **ICC** = intraclass correlation coefficient.

Keywords:
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